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A Decision Support System Using Text Mining Technology

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1. Introduction

With the rapid growth of the Internet, personal opinions regarding products (cameras, cars, books, movies, electronics, etc.) are spread very quickly on the Web. Effective use of this information would be beneficial to both consumers and manufacturers.

Blog, as one such information source has attracted much attention for these reasons. The personal opinions in blog, such as feedback surveys, bulletin boards, opinions and product reviews, make up an enormous amount of unstructured and/or semi-structured data. This information is factual, but in free format and mostly textual, which is very difficult to manage in a short time. Therefore, there is a great need for some techniques to handle such unstructured data effectively. Text mining, especially opinion mining, can play an important role in fulfilling this need [1].

Related works have mostly studied how to classify the opinions into bipolar orientation (i.e. either positive or negative opinion) [1] [2] [3]. These research results could not offer a comparison of the same kind of product and also could not recommend a suitable product according to the requirements of consumers.

To address the above question, this study proposes a decision support system based on the product opinion in blog. The target language of this system is Japanese. Instead of classifying the whole review, we analyzed the opinions on the phrase level. Another point in this study is the classification approach. It is necessary to classify the contents of blog into feature types(i.e. Price, Design or Function) and P/N (positive or negative) orientation. We implemented feature classification using a supervised approach (machine learning) and P/N Classification using an unsupervised approach (semantic orientation).

2. Proposed System

2.1 System Outline

The proposed system aims to collect and analyze opinions on a specific type of product automatically from blog, and then compare different products of this type, finally recommend a suitable product through interaction with the user.

An overview of this system is shown in Fig.1. The system is divided into two parts. Part 1 is information collection including step one to three. These steps will be repeated until the necessary information has been collected. Part 2 gives a recommendation to the user in accordance with the database information. The range of this paper is Part 1.

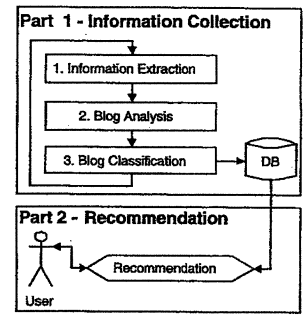


Fig. 1: Flow of the system

2.2 Information Extraction

The first step is Information Search. We used the Google Search Engine to get all the information on one type of product in the blog on the Internet. The Lynx, a text browser was used to get the contents of the pages returned by the search engine. Compared to the manual work, the system had a coverage of 87% on the extracted contents.

2.3 Blog Analysis

Blog analysis is used to extract the opinion phrases through morphological analysis and phrase analysis. There are four steps in this process. The first is Sentence Separation. All the sentences were separated by end punctuation (such as "！""．""。""？" etc). The second step is morphological analysis. We used a famous parser for Japanese called Chasen. Furthermore, we added an Unknown Corpus to Chasen to increase the precision of the analysis. The third step is preprocessing of opinion collection. It means filtering stop words (such as "Person", "Year" etc), and merging the words that have the same POS (a part of speech). Finally, based on the grammatical structure of Japanese and the results in related works [3], we designed the following templates to collect the opinion phrases. The obtained Recall was 71% and the Precision was 52%. The Precision was poor because general words were also extracted as opinion.

< noun + auxiliary word + adj/verb/noun >
< adj + noun/undefined/verb >
< noun + verb >

< noun + symbol + adj /verb/noun >

Except the above< adj >

2.4 Blog Classification

2.4.1 Feature Classification

Feature classification means to classify the phrase into Price(F1), Design(F2) or Function(F3) features. Both Naive Bayes and Support Vector Machines (SVM) are the well known algorithms for learning to classify text documents. From the comparative experiment, the precision of the two algorithms is very close, and the speed of the application is important in this system. Therefore, we chose Naive Bayes as the Feature Classification Method.

2.4.2 P/N Classification

P/N Classification aims to classify the phrases into positive or negative meanings using the SO (Semantic Orientation) value. We used and improved the SO-PMI (Semantic Orientation Using Pointwise Mutual Information) approach [2] to perform this classification.

The SO-PMI algorithm is employed to estimate the semantic orientation of a phrase by measuring the similarity of pairs of words or phrases. The Pointwise Mutual Information (PMI) between two words, word{1} and word{2} is defined as follows:

$$PMI(word_1, word_2) = log_2\left[\frac{p(word_1 \& word_2)}{p(word_1 \& word_2)}\right] \quad (1)$$

SO of a phrase is calculated here as follows:

$$SO(phrase) = PMI(phrase, \text{“}excellent\text{”}) - PMI(phrase, \text{“}poor\text{”}) \quad (2)$$

The reference words “ excellent ” and “ poor ” were chosen because, SO is positive when phrase is more strongly associated with “excellent” and negative when phrase is more strongly associated with “ poor ”.

Let hits(query) be the number of hits returned, the following estimate of SO can be derived from equations (1) and (2) with some minor algebraic manipulation.

$$SO(phrase) = log_2[A]$$

$$A = \frac{hits(phrase\ NEAR\text{“}excellent\text{”}) * hits(\text{“}poor\text{”})}{hits(phrase\ NEAR\text{“}poor\text{”}) * hits(\text{“}excellent\text{”})} \quad (3)$$

Following the characteristics of the Japanese, we improved the SO formula to the one shown in (4).

$$SO(phrase) = log_2[B] + threshold$$

$$B = \frac{hits(phrase\ NEAR\ good_basic) * hits(bad_basic)}{hits(phrase\ NEAR\ badbasic) * hits(good_basic)} \quad (4)$$

The reference words “ *excellent* ” and “ *poor* ” were replaced with “*good_basic*” and “*bad_basic*”. The “ *good_basic*” means the basic popular words for positive opinion in Japanese. And the “*bad_basic*” are the basic popular words for negative opinions in Japanese. The following words were used.

“ *good_basic* ” = “ よい | いい | 良い | 満足 | 良く | 好き | 優れて | お薦め | 利点 | 悪くない | いいかも | 嫌いじゃない | 傑作 | 大好き | 良かった | 満たしている ”

“ *bad_basic* ” = “ 悪い | 嫌 | 不満 | 嫌い | 不安 | 不良 | 最悪 | 悪く | 欠点 | 満足ではない | 好きではない | 不具合 | 欠陥 | 悪質 | 不快 | 気に入らない | よくない | 認めない | 許せない | あまり | 良くない | 満たしていない ”

In a preliminary experiment, we used zero as a threshold. The result of this trial was very unsatisfying. The precision of positive opinions was 21.02%, and the precision of negative opinions was 21.69%. Therefore, we decided to setup our own baseline. The threshold used was the following:

$$threshold = log_2\left[\frac{hits(bad_basic)}{hits(good_basic)}\right] * (-0.6) \quad (5)$$

After this improvement, the precision of positive opinions was 70.91%, which was quite good. The precision of negative opinions was only 33.75%. The cause is that the analysis of Japanese negative expressions is still insufficient.

3. Conclusions

This paper proposed the entire structure of a decision support system based on the product opinions in blog. And the detailed approach in every process about system's Part 1 (Information Collection) was presented. Especially, we improved the SO-PMI approach to perform P/N classification, and then achieved better precision.

In the future, we will continue to improve the Blog analysis and Blog Classification. In the process of Blog analysis, we should study how to distinguish between opinion phrases and general words. About P/N classification, we will work on analyzing the negative expressions and adjusting the threshold. Furthermore, we also want to achieve Part 2 (Recommendation) to realize the final purpose of this system.

References

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